

Answers to NJC H2 Biology Preliminary Paper 2

Section A

1 Fig. 1.1 is a transmission electron micrograph of a mammalian cell.

(a) Complete Table 1.1 to:

- name in full, structures **A**, **B**, and **C**.
- outline how each structure functions to contribute to its specific role in the mammalian cell.

Table 1.1

<i>structure</i>	<i>name of structure</i>	<i>function of structure within mammalian cell</i>
A	<i>A: nucleus / (eu)chromatin</i> <i>R: heterochromatin / chromosome / DNA</i>	• <i>Contain gene(s)/ genetic information/ genetic material / DNA that codes for proteins/ polypeptide</i> • <i>Site for transcription (occurring) leading to mRNA synthesis</i>
B	<i>mitochondrion</i> <i>R: mitochondria</i>	• <i>Provides / synthesizes / produces / makes ATP for protein synthesis / exocytosis</i>
C	<i>Rough endoplasmic reticulum</i>	• <i>Synthesis / modification / processing / transport, of protein / polypeptide</i> <i>A: translation</i>

[3]

Table 1.2 below represents the ionic concentrations inside and outside a typical mammalian cell.

(b) Explain how the intracellular concentration of Na^+ is maintained.

[1]

- *Sodium-potassium pumps pump out excess Na^+ thus requires ATP for the active transport of this ion.*

Cholera, a disease caused by a bacterial infection of *Vibrio cholerae*, results in severe diarrhoea leading to dehydration. A toxin released by the bacterium causes the release of chloride ions (Cl^-) from cells lining the small intestines into the lumen and inhibits the uptake of sodium ions (Na^+) by these cells. Under normal circumstances, the Cl^- intracellular concentration is maintained due to the natural electrical negativity of the cell.

(c) By comparing normal and diarrhoeal situations, explain how this disruption of cellular ion concentrations would result in extreme dehydration. [4]

- *Under normal conditions, the chloride ions remain in the cells lining the lumen of the small intestine → creating an overall negative charge within the cells that drives positively charged sodium ions into the cells*
- *This creates a hyperosmotic condition in the cells/ lower water potential in cells (or content of the cell would contain more solutes than extracellular fluid) → Thus resulting in water moving into the cells by osmosis.*
- *When suffering from diarrhoea, there is increased concentration of Cl^- and Na^+ in the lumen of the small intestine.*
- *Elevated concentrations of these two ions results in the lumen of the small intestines becoming hyperosmotic/ having a lower water potential compared to the cells lining the intestines.*
- *This in turn causes large amounts of water through osmosis into the lumen of the intestine from the cells and blood stream → Water passes out of the body as diarrhoea, leaving the body dehydrated.*

(d) One of the antibiotics used in the treatment of cholera is tetracycline. Tetracycline causes the death of the pathogen by binding to ribosomes and inhibiting protein synthesis.

(i) Suggest two ways in which tetracycline acts at ribosomes to inhibit protein synthesis. [2]

- *Binding of tRNA is prevented → no codon-anticodon binding and peptide bond formation is prevented*
- *mRNA attachment is prevented → ribosome movement along the mRNA is hindered / prevented*
- *Inhibits the association of large and small subunits → ribosome movement along the mRNA is hindered / prevented*

(ii) Tetracycline does not harm mammalian cells. Suggest an explanation for this.
[1]

- *Cell surface membrane of mammalian cell is impermeable to tetracycline*
- *Possibility of tetracycline to be broken down by enzymes / be degraded before entry into the cell*
- *Tetracycline targets prokaryotic ribosome and the structure of eukaryotic ribosome is different → difference in ribosomal structure*

[Total: 11]

- 2 Meselson and Stahl performed the following experiment with the bacterium *Escherichia coli* (*E. coli*). They grew cells for several generations in the presence of the heavy isotope of nitrogen, ^{15}N . All the DNA of the progeny that were produced from the initial small inoculum was labelled with ^{15}N . The ^{15}N -labelled cells were then placed in ^{14}N -containing medium. The DNA produced in subsequent generations was extracted. The DNA from the different generations was compared using the technique of differential centrifugation using a caesium chloride density gradient to separate the DNA containing ^{15}N from DNA containing ^{14}N , as ^{15}N labelled DNA is more dense than ordinary ^{14}N DNA. The position of the ^{15}N and ^{14}N DNA absorbance was determined in ultraviolet light at 260 nm. The results obtained are shown in Fig. 2.1.

- (a) (i) With reference to Fig. 2.1, do you consider the replication of DNA in *E. coli* as conservative or semi-conservative? Give reasons for your answer. [3]

- *It is semi-conservative, because after one generation in ^{14}N , the DNA had a density intermediate between that containing ^{15}N and that with ^{14}N .*
- *After 2 generations in ^{14}N , the DNA was of two types: one with an intermediate density and another with a density corresponding to ^{14}N DNA.*
- *This shows that the DNA molecule unzips and each original strand acts as a template for the synthesis of new strand. Daughter DNA contains one old and one new strand. If it is conservative, in each generation there would always present DNA that has density similar to the parent DNA.*

- (ii) On Fig. 2.1, draw the expected position of the ^{15}N and ^{14}N DNA absorbance after three generations in ^{14}N . [1]

- *Two peaks: A higher peak at ^{14}N and a smaller peak at ^{14}N - ^{15}N sedimentation*

- (iii) Explain the importance of hydrogen bonding in DNA structure. [2]

- *Hydrogen bonds hold the two polynucleotides / strands / chains together via complementary base pairing*
- *Many hydrogen bonds give stability as DNA is a stable molecule*
- *Hydrogen bonds can be broken for the process of DNA replication/ transcription to take place*
- *Reference made to structure of DNA as double helix*

- (iv) The enzyme that catalyzes the replication of DNA checks for errors in the process and corrects them. Suggest why checking for errors and correcting them is necessary. [1]

- *Errors are mutations and the presence of mutations may lead to production of altered proteins / impaired / loss of function*
- *Altered proteins may lead to different antigens on the surface of cells, so cells will be rejected by immune system*
- *Impaired coordination of cells → i.e. cancerous cells whereby there is uncontrolled cell division / cell replication / cell growth OR lack of contact inhibition / no apoptosis / proto-oncogenes*

A group of scientists has discovered that a change in the chromatin structure of mammalian cell brought about by Protein A is associated with increased gene expression of several genes involved in the suppression of tumour growth. Fig. 2.2 shows the overall survival (OS) percentage of patients in the absence and presence of Protein A.

- (b) (i) With reference to Fig. 2.2, comment on the overall survival percentage of patients in the presence of Protein A with that in the absence of Protein A, and explain for its differences. [4]

- *Quote data for both absence and presence of Protein A*
- *Protein A is involved in histone acetylation where the acetyl groups are attached to positively charged lysine amino acids in histone tails*
- *When the histone tails of a nucleosome are acetylated, their positive charges are neutralized*
- *Results in the chromatin having a less compact structure leading to an increase in the rate of transcription of tumor suppressor gene*

- (ii) Explain how the presence of transcription factors causes an increase in the expression of the gene involved in suppression of tumour growth. [3]

- *Compulsory marking point: Activator proteins (specific transcription factors) bind to distal control elements grouped as an enhancer in the DNA.*
- *DNA-bending protein binds to the DNA and brings the bound activators closer to the promoter.*
- *Enables the enhancer to influence a promoter hundreds or even thousands of nucleotides away. Other transcription factors, mediator proteins and RNA polymerase are nearby in the cell's nucleus.*
- *The activators bind to certain general transcription factors and mediator proteins. These multiple protein-protein interactions help position and increase the rate of assembly of the transcription initiation complex on the promoter for RNA synthesis to begin.*

- (iii) Suggest why the change in chromatin structure, in the presence of Protein A, does not occur over large area of the chromatin [1]

Reason 1:

- *Only a small proportion of genes are to be expressed for tumor suppression*
- *Thus genes that code for proteins that are not required in that particular cell type will have their transcription repressed instead*

Reason 2:

- *Size of eukaryotic genome is very large with largely non-coding regions*
- *thus histone acetylation (changes to the chromatin structure) do not occur at these regions as no gene expression is required*

[Total: 15]

- 3 Human Immunodeficiency Virus (HIV) is the culprit of Acquired Immunodeficiency Syndrome (AIDS) which, till now, has no full cure.

Figure 3.1 shows the reproductive cycle of HIV.

- (a) Name structure A and B. [2]

A: gp120

B: CD4 receptor

- (b) Describe the events occurring in stage C leading to formation of structure A on the host cell's plasma membrane. [3]

- Polypeptides translated by ribosomes enter the lumen of rough endoplasmic reticulum for proper folding.*
- Proteins are packaged in transport vesicles which bud off RER and fuse with cis face of golgi apparatus in which glycosylation occurs.*
- Glycoprotein is inserted into the membrane of secretory vesicles when they bud off the trans face of golgi apparatus and fuse with the plasma membrane. The glycoprotein/structure A will be incorporated in the plasma membrane.*

- (c) After contraction of HIV, the initial symptoms are followed by a stage called clinical latency when no symptoms are observed for a few years. With reference to the reproductive features of HIV, explain the cause of clinical latency. [1]

- The reverse transcribed HIV DNA is integrated into the host chromosome as provirus. Provirus remains silent/ transcriptionally inactive for years, causing no symptoms.*

- (d) Flu is caused by influenza viruses.

- i. Distinguish the genome replication between HIV and influenza virus. [3]

<i>Feature</i>	<i>HIV</i>	<i>Influenza virus</i>
<i>Beginning material</i>	<i>Single stranded positive/sense RNA</i>	<i>Single stranded negative/anti-sense RNA</i>
<i>Viral enzymes involved</i>	<i>Reverse transcriptase synthesize double stranded cDNA from ss RNA;</i> <i>Integrase incorporate the viral cDNA as provirus into host chromosome</i>	<i>RNA-dependent RNA polymerase synthesizes the complementary sense strand followed by anti-sense strand</i> <i>The viral genome is not integrated into the host chromosome</i>

<i>Time taken to replicate the genome</i>	<i>Provirus remains silent for years. It will be transcribed to form the positive ssRNA and mRNA when provirus is activated to produce new HIV virion.</i>	<i>Fast and immediate.</i>
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Examiner's comments: *Not sufficient information is given in this question for marks to be awarded. Enzymes should also be stated when mentioning processes.*

- ii. Every year, new flu vaccines are developed to ensure their efficacy. Explain the frequent need for new flu vaccines. [2]
- Influenza viruses undergo antigenic drift frequently due to high mutation rate during its genome replication leading to change in conformation of haemagglutinin and neuraminidase on its viral envelope*
 - The lack of proof reading ability of RNA-dependent RNA polymerase as well as the fast replication involving two rounds of complementary RNA strand formation account for the high mutation rate.*

[Total: 10]

4 The arrow in Fig. 4.1 shows the direction of an impulse in part of a nervous system.

(a) With reference to Fig. 4.1, describe how transmission of impulse is believed to take place. [4]

- *Arrival of nerve impulse at synaptic knob depolarises presynaptic membrane and increases permeability to Ca^{2+} , causing influx of Ca^{2+} .*
- *Synaptic vesicles move, fuse with presynaptic membrane to discharge neurotransmitters by exocytosis.*
- *Neurotransmitter diffuses across cleft to attach to specific receptor sites on post-synaptic membrane. Ligand gated ion channels opens up to allow influx of Na^{+} .*
- *This bring about local depolarization which when exceeds threshold, an action potential is triggered in post-synaptic neurone.*

(b) When summation of EPSP and IPSP exceeds threshold, depolarisation occurs. Explain why the magnitude of the signal initiated at the axon hillock is always at constant voltage. [2]

- *When threshold potential is reached, all the voltage sensitive sodium ion channels will open, resulting in the influx of Na^{+} .*
- *When +50mv is reached, all the sodium ion channels will closed*
- *It is an all-or-none event/Axon hillock naming*

(c) Suggest the purpose of the refractory period during an action potential. [2]

- *Cause/explanation of refractory period*
- *Prevent over fatigue of the neuron*
- *Prevent back stimulation of the axon.*

In a nerve muscle preparation, a novel drug called AWZ with a molecular structure resembling acetylcholine was applied to the cells. The investigation showed that the drug significantly increases muscular contractions. However, it was found that the drug, when applied to either one kind of cells directly, did not affect the axon nor muscle cells.

(d) Suggest, in detail, one possible way in which the drug, AWZ, could cause the increased muscle contractions. [4]

- *Shape similar to acetylcholine. Serve as competitive inhibitor to acetylcholinesterase*
- *Competes with acetylcholine for the active site*
- *Acetylcholine unable to be broken down and continuously bind to receptors on post synaptic membrane.*
- *Prolonged binding to receptors at post synaptic membrane*
- *leads to increased frequency of depolarization at the post-synaptic neuron.*

[Total: 12]

- 5 (a) Pituitary dwarfism is a sex-linked inherited condition in humans in which affected individuals have very short limbs. The family tree, in Fig. 5.1, shows part of one affected family.

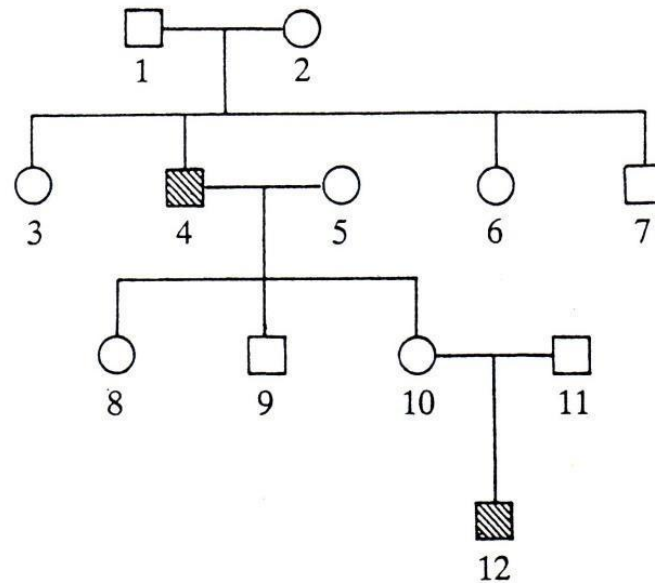


Fig. 5.1

- (i) Explain what is meant by 'sex-linked inherited condition'. [1]

- A character that is controlled by a gene situated on the X-chromosome or Y chromosome, although X being more common.

- (ii) With reference to Fig 5.1, explain the mode of inheritance using two pieces of evidences. [4]

1st Piece of evidence:

- If the allele for dwarfism is dominant, the **affected male, 4**, who has only one X chromosome should pass this allele to all his daughters, who should then be affected. [Ref to all daughters should be affected]
- However, his daughters, 8 and 10 are normal. Hence allele for this condition is not dominant but recessive.

OR

- Male 4 receives its only X chromosome carrying the allele for dwarfism from his mother 2 (since he would have obtained Y chromosome from his dad)
- Hence his mother 2 is a heterozygote with an allele for normal limb and an allele for dwarfism
- Since the **mother of male 4 is normal**, the allele for dwarfism must be recessive to the allele for normal limb.

A series of breeding experiments were carried out with a certain species of mice. A cross between pure breeding males with red eyes and bent tails, and females with black eyes and straight tails produced offspring all with red eyes and bent tails. The same results were obtained when the cross was carried out again but with the sexes reversed.

(b) Suggest a reason for carrying out the cross with the sexes reversed. [2]

- To check whether the genes are **sex-linked**. [Answers stating linked genes were rejected]
- If sex-linkage involved as the mode of inheritance, then offspring **phenotypic ratio** will be different.

In another crossing, female offspring from the first experiment were then crossed with black eyes and straight tails males to obtain the following:

185 red eye, bent tail

153 red eye, straight tail

160 black eye, bent tail

188 purple eye, straight tail

- (c) Using the given formula and part of the table of chi-square values as shown in Table 5.1, determine if the null hypothesis for the eye color allele and tail shape allele can be accepted in the space below. [3]

$$\chi^2 = \sum \frac{(o - e)^2}{e}$$

Part of the table of chi-square values is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

$$\chi^2 = 1.06 + 2.00 + 0.77 + 1.59 = 5.42; \text{ Degree of freedom} = 4 - 1 = 3; \text{ At } P = 0.05, \chi^2 = 7.81$$

Since $5.42 < 7.81$, there is no significant difference between observed and expected and is due to chance. Therefore accept null hypothesis.

[1] - Given to correct calculation of 5.42,

[1] - Given to stating Degree of freedom and or P and calculated χ^2 Value. [Students are required to state all 3 items!]

[1] - State the conclusion of the hypothesis. Reject or accept.

[Total: 10]

6 Fig. 6.1 illustrates what happens on thylakoid membrane during daytime.

(a)

(i) Describe how electron transport chain leads to ATP production. [3]

- *When electrons are transferred through a series of electron carriers arranged in a progressively lower energy fashion, energy is released during each transfer*
- *The energy is use to actively transport H^+ against its concentration gradient from stroma to the thylakoid space, creating a electrochemical H^+ gradient.*
- *Chemiosmosis occurs when the H^+ flows down its concentration gradient via facilitated diffusion through ATP synthase, driving the synthesis of ATP from ADP and P_i in the stroma.*

(ii) Compare the initial electron donor and final electron acceptor between oxidative phosphorylation and photophosphorylation. [2]

	<i>Oxidative phosphorylation</i>	<i>Photophosphorylation</i>
<i>Initial electron donor</i>	<i>$NADH_2$ and $FADH_2$</i>	<i>H_2O</i>
<i>Final electron acceptor</i>	<i>O_2</i>	<i>$NADP^+$</i>

(b) A point mutation results in dysfunctional $NADP^+$ reductase. State and explain how the concentration of Ribulose 1,5 – Bisphosphate in the chloroplasts would be affected. [3]

- *$NADP^+$ cannot be reduced to $NADPH$ which is required as a reducing power for the conversion of 1,3-bisphosphoglycerate to glyceraldehydes 3-phosphate.*
- *The subsequent regeneration of Ribulose 1,5 – Bisphosphate is prevented while carbon fixation continues to occur,*
- *depleting the supply of Ribulose 1,5 – Bisphosphate, causing its concentration to drop.*

(c) Glyceraldehyde 3-phosphate exits the Calvin cycle for production of glucose, which will be used for cell wall formation. Explain how the structure of cellulose allows it to serve its function. [4]

- *About 10,000 β -glucose units with alternating units rotated 180° in space are linked by $\beta(1,4)$ glycosidic bonds to form long and linear chains.*
- *Large number of H-bonds / extensive cross-links is formed between the linear cellulose chains due to the large number of β -glucose units with protruding $-OH$ groups in each chain.*

- *These cross-linked chains are associated to form microfibrils, which are then meshed into a macrofibril matrix .*
- *The resultant great tensile strength & insolubility in water allows cellulose to be the main component of plant cell wall, providing structural support to plant cells. The porous matrix allows free passage of water molecules.*

[Total: 12]

- 7 The greenish warbler, *Phylloscopus trochiloides*, is a species of small bird that originated in northern India, on the southern edge of the Himalayan mountain range.

(a) Describe the principles which scientists use to classify organisms into taxonomic groups. [2]

- *Smaller groups within big groups → hierarchical (compulsory marking point)*
- *Based upon similarities / features in common*
- *Evolutionary relationships / common ancestry / phylogenetics*

Fig. 7.1 shows a greenish warbler.

Thousands of years ago, populations of the greenish warbler spread around the western and eastern edges of the Himalayan mountain range to establish themselves in north-eastern Europe and Siberia.

- A gradual change in characteristics occurred in these populations, leading to different forms of the greenish warbler.
- One example of gradual change is in the song of the male warbler, which is very distinctive and is used in mating behavior.
- When greenish warblers from north-eastern Europe meet those from Siberia, no mating takes place.
- The greenish warblers from north-eastern Europe and Siberia are now considered to be two separate species.

Fig. 7.2 shows the spread of the greenish warbler.

(b) State the likely isolating mechanism taking place in populations of the greenish warbler. [1]

- *Geographical barrier / isolation*

(c) Explain how the process of speciation occurred in the greenish warbler populations. [4]

- *Breeding/gene flow between populations is prevented*
- *Gene mutations occurring – to make reference to at least 1 type of mutation that leads to change in protein sequence*
- *Different selection pressures/different (environmental) conditions*
- *Genetic change, whereby different alleles selected for/change in allele frequency/change in gene pool/advantageous alleles passed on*
- *Presence of genetic drift*
- *Do not recognize the song → (ultimately) cannot interbreed/reproductively isolated*
- *Eventually, allopatric speciation*

An investigation was carried out to study the relationships between the different species of warbler using DNA analysis. The base sequences of a region of mitochondrial DNA from the four species were compared and results are shown in Table 7.1.

Table 7.1

	<i>P. laetus</i>			
<i>P. laetus</i>		<i>P. laurae</i>		
<i>P. laurae</i>	12.1		<i>P. ruficapilla</i>	
<i>P. ruficapilla</i>	16.7	15.0		<i>P. affinis</i>
<i>P. affinis</i>	11.3	8.9	13.2	

- (d) Researchers have put forward the hypothesis that the three species, *P. laetus*, *P. laurae* and *P. ruficapilla*, have originated from three separate events in which a few individuals of *P. affinis* spread directly from Cuba to three different places. The smaller the number, the smaller the differences between the base sequences of the two species.

Explain how the results in Table 7.1 support the researchers' hypothesis. [2]

- *P. laetus*, *P. laurae* and *P. ruficapilla* have smaller differences with *P. affinis* → Thus, are more closely related to *P. affinis* than to each other
- [Quote figures]: For example the number of base differences between *P. laurae* and *P. affinis* is 8.9, as compared to *P. ruficapilla* and *P. laurae* which is 15.0.
- Comment about figures for *P. laurae* with *P. laetus* with reference to different times of separation

- (e) It was observed that the warblers have remained at approximately the same frequency for many generations. Suggest an explanation for this phenomenon. [1]

- Migration of birds from mainland
- Presence of homozygote / heterozygote advantage
- Stabilizing selection
- Selection pressure remains the same.

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 6 (a)** Describe homologous chromosomes and distinguish between their behaviors in mitosis and the first division of meiosis. [10]

Similarities:

- *Same number of genes;*
- *Carry the same genes / carry code for the same characteristics;*
- *Same relative position / loci along the chromosome;*
- *Same length/ size/ shape;*
- *Position of the centromere is the same;*

Differences:

- *Different alleles/ carry genes of an alternative kind;*
- *Each originates from different parents;*
- *Different sequence of bases / nucleotides*

STAGE	MITOSIS	FIRST DIVISION OF MEIOSIS
<i>Prophase</i>	<i>No synapsis / no pairing of homologous chromosomes</i> <i>No crossing over / no chiasma formation</i>	<i>Synapsis occurs / homologous chromosomes pair up to form bivalents</i> <i>Crossing over occurs between non-sister chromatids of homologous pair at chiasmata.</i>
<i>Metaphase</i>	<i>Chromosomes arrange lengthwise / on the same plane / in a single row on the equator of spindle.</i> <i>No independent assortment since sister chromatids are identical.</i>	<i>Homologous chromosomes lie side-by-side / equidistant on either side of the equator.</i> <i>Independent assortment of alleles occur.</i>

Anaphase	Centromere of each chromosome divide Sister chromatids separate OR Separating chromatids are identical	Centromeres do not divide Homologous chromosomes separate OR Separating chromosomes are not identical.
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(b) With reference to specific genes, discuss how point mutations and chromosomal mutations can lead to cancer. [10]

- *Proto-oncogenes such as Ras gene codes for Ras protein which is involved in promoting mitosis in the presence of signaling molecules such as epidermal growth factor.*
- *Gain-of-function mutations change the proto-oncogene to oncogene, causing production of hyperactive Ras protein or over-production of Ras protein.*
- *Base pair substitutions in the coding sequence can results in missense mutations, coding different amino acids and resulting in altered three dimensional conformation of the protein. For instance, the GTPase active site could be altered and is no longer able to catalyse hydrolysis of GTP bound to Ras protein, making it perpetually active / hyperactive.*
- *Base pair substitutions in the regulatory sequences such as promoter and enhancer of the Ras gene can transform the promoter into a more powerful regulator of gene transcription, resulting in large quantity of Ras protein synthesized.*
- *Chromosomal translocation can cause Ras gene next to a stronger promoter, resulting in over-production of Ras protein.*
- *Aneuploidy as a result of non-disjunction during mitosis can lead to more than 2 copies of the Ras gene present in a cell, resulting in over-production of Ras protein also.*
- *Tumor suppressor genes such as p53 gene codes for p53 protein which is a transcription factor switching on genes whose products are involved in repairing DNA damage or triggering apoptosis should the damage is beyond repair.*
- *Point mutations and chromosomal mutations can lead to loss-of-function mutation in p53 gene, leading to nonfunctional p53 protein or inadequate amount of p53 protein.*
- *Base pair substitutions in the coding sequence of p53 gene can lead to missense mutations, coding for a different amino acid. The tertiary structure of p53 protein is altered, rendering it dysfunctional.*

- *Base pair substitution in the regulatory sequence such as promoter or enhancer of p53 gene can lead to a non-functional promoter or enhancer not recognizable by transcription factors and RNA polymerase, hence leading to transcriptional inactivation. No or insufficient p53 protein is produced.*
- *A chromosomal translocation of p53 gene next to a non-promoter region such as centromere or other repeated sequence cause the gene to be silent / no expressed.*
- *Aneuploidy as a result of non-disjunction during mitosis can lead to only 1 copy of p53 gene present, increasing the likelihood of losing functional p53 proteins as a mutant allele will not be masked by the other normal allele.*
- *Chomosomal deletion may also result in loss of heterozyosity. One less copy of p53 gene is present in the genome. Mutation in the only allele will directly cause loss of functional p53 proteins in the cell.*
- *When both p53 alleles undergo loss-of-function mutation, no functional p53 proteins are produced to arrest cell cycle when there is severe DNA damage, causing genetic instability.*
- *When mutations accumulate in the genome, especially mutations in other tumor suppressor genes and proto-oncogenes occur, cells will undergo mitosis uncontrollably, leading to cancer.*

7 (a) Describe how the effects of glucagon are achieved in the liver cell. [10]

1. *Ligand-receptor interactions:*

- *Hormone glucagon binds to a G protein-coupled receptor (GPCR) present on the cell surface membrane of liver cells*
- *Ligand-receptor interaction results in the formation of a hormone-receptor complex*
- *The hormone glucagon is the first messenger that triggers the subsequent intracellular signaling events in the target cell*

2. *Signal Transduction:*

- *Binding of glucagon triggers a conformation change in the GPCR*
- *providing a signal that activates the G protein (found on the cytoplasmic side of the membrane) by replacing the GDP to a GTP*
- *The activated G protein diffuses along the plasma membrane to bind to and activate adenylyl cyclase which in turn catalyses the conversion of many cyclic AMP (cAMP) molecules from ATP*
- *cAMP acts as a second messenger (within the liver cell) to activate enzymes such as protein kinase A*
- *Protein kinase A activate other enzymes by phosphorylating them triggering a phosphorylation cascade that serves to amplify the initial signal where the binding of a single glucagon molecule results in a strong response inside the cell/ number of enzyme molecules activated each step exceeds the preceding step*

3. *Cellular response:*

- *(Phosphorylation cascade) finally leads to activation of glycogen phosphorylase which is needed for the breakdown of glycogen stored in liver cells to glucose molecules*
- *thereby increasing the availability of glucose for cellular respiration / bring blood glucose levels back to the norm*

- (b) Discuss why the eukaryotic genome is considered to be more complex than the prokaryotic genome. [10]

Prokaryotic Genome	Eukaryotic Genome
<i>Smaller number of bases</i> ○ <i>Smaller number of genes / coding regions</i>	<i>Larger number of bases</i> ○ <i>Larger number of genes / coding regions</i>
<i>Single chromosome</i> ○ <i>Usually haploid (i.e. only 1 set of chromosomes)</i> ○ <i>No pairing of homologous chromosomes possible, hence no genetic recombination as a means of giving rise to genetic diversity</i>	<i>Multiple chromosomes</i> ○ <i>Usually in diploid or higher ploidy levels</i> ○ <i>Pairing of homologous chromosomes, allows for genetic recombination during meiosis, giving rise to genetic diversity</i>
<i>Absence of / very little intergenic regions</i> ○ <i>Most of the genome is coding regions, hence little regulatory elements possible to be present</i>	<i>Presence of intergenic regions</i> ○ <i>Presence of repetitive DNA, many whose functions are yet unknown and may play a role in gene regulation</i>
<i>Polycistronic genes / operons present</i> ○ <i>Allows coordinated control of metabolic enzymes involved in the same pathway / all products of operons produced in the same amount</i> ○ <i>However this coordinated control results in fewer possible metabolic pathways for a given substrate as all substrate will be fed into that one single pathway</i>	<i>Monocistronic genes</i> ○ <i>Individual enzymes are controlled individually by separate promoters</i> ○ <i>Can be produced in different amounts or not at all</i> ○ <i>Hence allowing for partitioning of metabolic intermediates to different metabolic pathways to suit the cells' needs</i>
<i>Absence of control elements</i> ○ <i>Allows only single-step control of gene expression, i.e. dependent on promoter strength and operator state</i>	<i>Presence of control elements</i> ○ <i>Fine tuned control of gene expression, allows for different "levels" of expression of a particular gene depending on the activators and repressors present in the cell at a specific time</i>
<i>Absence of introns</i>	<i>Presence of introns</i>

○ <i>No alternative splicing possible</i> ○ <i>Hence only one possible protein product per gene</i>	○ <i>Allows alternative splicing</i> ○ <i>Presence of introns allows for choice of splice sites, giving rise to varying mature mRNA molecules and hence multiple possible protein variants from the same gene</i>
<i>No association with histones</i> ○ <i>Association instead with proteins similar to (but not the same as) histones</i> ○ <i>Does not achieve same level of condensation complexity as eukaryote</i>	<i>Association with histones</i> ○ <i>Allows for increased structural complexity / folding to higher degree of condensation</i> ○ <i>Control of rate of gene transcription by allowing for increased DNA condensation/conversion between euchromatin and heterochromatin states</i>
<i>Gene amplification does not occur</i> ○ <i>No mechanism for increasing copy numbers of genes</i>	<i>Gene amplification is possible</i> ○ <i>Increased copy number of genes allows for increased transcriptional levels</i>
<i>No nuclear membrane</i> ○ <i>Simultaneous transcription and translation</i> ○ <i>Hence fewer levels of control of gene expression</i>	<i>Enclosed by nuclear membrane</i> ○ <i>No simultaneous transcription and translation</i> ○ <i>Allowing for more levels of control of gene expression</i>

--- The End ---